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Policy capacities for mission-oriented innovation policy: A case study of UKRI and the industrial strategy challenge fund

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ABSTRACT

A new generation of transformative innovation policies, including a reimagined 'mission' approach, centre around tackling societal challenges. Their emphasis on directionality, coordination of different actors and policy mixes goes beyond standard science, technology, and innovation (STI), engaging wider policy and other domains. Whilst missions have become steadily popular and there is growing research on their design, there is much to understand about how such models are implemented in practice. A relative gap in the literature is how science, technology and innovation (STI) organisations develop the policy capacities and capabilities to deliver missions in context. This article provides a case study on how UK Research and Innovation (UKRI) fared whilst implementing the Industrial Strategy Challenge Fund (ISCF). It finds that the policy capacities for missions were not fully present within UKRI and notes several tensions as the organisation worked to develop its capabilities for implementing missions. The paper makes recommendations for how funders can enhance those policy capacities and suggestions for how mission-oriented innovation policy analysis might be developed.

1. Introduction

The focus of this article is on mission-oriented innovation policies and their implementation, with a particular focus on how science, technology, and innovation (STI) bureaucracies that lead public investment in R&I are developing policy capacities to design and capabilities to implement new measures. The policy context is the UK which boasts one of the best research and innovation (R&I) systems internationally, with successive governments positioning research and development (R&D) as fundamental to fixing societal problems.[1] However, a key question is whether the UK is developing the policy capacities and capabilities to mobilise the full range of its knowledge and innovation assets and deliver workable solutions to problems including the impacts of climate change or how we live healthy lives.

UK Research and Innovation (UKRI) is the primary agency for public investment in science, research and innovation. It was established in 2018, which coincided with the UK's experiment with a more directional approach to R&I via the Industrial Strategy Challenge Fund (ISCF),[2] which UKRI led. Twenty-four goal-oriented 'Challenges' were launched under ISCF, each led by a Challenge Director from industry. Four years on, when this research began, the ISCF had been through several changes as it incorporated learning to date and adapted to developments in the wider political and policy landscape at that time. This included the government's *Build Back Better* framework (HMT, 2021), the Innovation Strategy (BEIS, 2021a) which supplanted the 2017 Industrial Strategy (BEIS, 2017), and the 2022 White Paper *Levelling-up the United Kingdom* (HMG, 2022). These agendas and initiatives impacted on overall innovation policy, and the institutional architecture for its delivery, which includes UKRI and a new Advanced Research and Invention Agency (ARIA), founded in early 2022 (BEIS, 2021b; HoL, 2021).

The UK's experiment lines up with a broader upswing of interest in transformative innovation policies (Schot and Steinmueller, 2018; Weber and Rohracher, 2012), including mission-oriented policies (Foray, 2018; Mazzucato, 2018a; Mazzucato, 2021), as governments

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[1] Former Prime Minister Rishi Sunak committed to cement the UK's position as a "science superpower" in the previous government's Science and Technology Framework (DSIT, 2023). A new Labour Government in the UK promised several measures of support of science, technology and innovation included in its manifesto, including a new industrial strategy and the end to short funding cycles for Labour Party (2024). A new Labour government in the UK is also committing to introduce mission-driven government, which will no doubt include the R&I system in delivering its ambitions.

[2] The ISCF was aligned to the 2017 Industrial Strategy. Its launch preceded the Industrial Strategy by 6–12 months (see timeline below) though the two were closely entwined in the planning stages of the Industrial Strategy with ISCF being positioned as a significant funding vehicle for the strategy.

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seek to direct innovation, economic activities and societal initiatives towards addressing complex problems. These policies emphasise the dynamic nature of societal problems and potential solutions, thereby pushing innovation policy actors to think and act systemically (Wittmann et al., 2024; Larrue, 2021; Grillitsch et al., 2019).

The literature on mission-oriented innovation explores types and frameworks to guide their design and implementation (see Larrue, 2019; Wittmann et al., 2020; Biegelbauer et al., 2020; Breitinger et al., 2021). However, what is less well researched and understood, is how mission approaches are being implemented in context, and more specifically, how organisations leading mission-oriented approaches adapt their existing policy capacities and capabilities as a result (Wittmann et al., 2021). This includes the ways in which existing capacities are adjusted to support directionality, a defining feature of transformative approaches (Bergek et al., 2023), along with activating and facilitating multiple processes of 'horizontal' and 'vertical' coordination to animate and support the ambition and demand aspects of system-wide change (Boon and Edler, 2018). There are notable studies, which have explored the challenges faced by STI organisations as they incorporate transformative innovation thinking and practice, and we draw upon those here (Diercks, 2019; Ulmanen et al., 2022; Chataway et al., 2017). However, these are modest in number, and this remains a relatively under-researched theme in the overall transformative innovation literature.

This paper specifically focuses on challenge-led missions and seeks to contribute to this field of practical studies. Our approach is nested in a growing innovation policy capacities and capabilities literature (e.g., Borrás, 2019; Borrás and Edler, 2020; Karo and Kattel, 2018; Kattel and Mazzucato, 2018). The main research question is how STI institutions leading challenge-led missions or programmes, in this case UKRI, develop policy capacities and capabilities for successful implementation in the process of delivering a challenge-led mission approach, and what, if any, challenges they face. The case study involves documentary review and interviews with senior leaders in UKRI including Challenge Directors appointed from industry to lead specific Challenge areas, in recognition of the role along with senior public administrators in Innovate UK, individual Research Councils and the Department for Business, Energy and Industrial Strategy (BEIS). An abductive, qualitative approach allows the empirical findings to be brought into dialogue with the literatures on missions and capacities, enabling us to explore how policy capacities for missions were being interpreted and evolving in practice.

To answer our research question, the sections that follow include a literature review which explores mission-oriented policies and their implementation, and policy capacities in the public sector. We draw on these to propose a framework of capacities for challenge-led mission approaches. The third section introduces the case and methodological considerations, followed by a fourth section on the findings which takes a narrative approach to illustrate how key figures across UKRI interpreted the role of ISCF and the capacities needed for challenge-led missions, against a backdrop of institutional change. The findings indicate that the full range of capacities we might expect was not in place and several tensions were present, between individual and institutional actors. This was particularly evident between officers or civil servants used to traditional practices or tasks deployed by the research councils or Innovate UK prior to UKRI being established, and Challenge Directors or officers specifically brought in to introduce and undertake new tasks towards the implementation of missions. The final section presents a concluding discussion of the findings and the implications for theory and policy.

2. Literature review: conceptualising policy capacities for mission-oriented innovation

2.1. Types and characteristics of mission-oriented innovation policy

Mission-oriented innovation policies form part of a broader paradigmatic shift towards more transformative innovation policies as governments seek to address complex societal problems and guide development towards broader societal transitions. This shift is underpinned by two main emerging literatures on 'transformative innovation policies' (Schot and Steinmueller, 2018; Haddad et al., 2022;) and 'mission-oriented' policy frameworks (Mazzucato, 2018a; Foray, 2018; Hekkert et al., 2020). These policies acknowledge the shortcomings of conventional science, technology, and innovation (STI) policy models for their lack of directionality and coordination, and instead offer an approach for reaching across policy domains, institutions, and sectors, thereby pushing innovation policy actors to think and act systemically (Diercks et al., 2019; Grillitsch et al., 2019).

The emphasis here is on state-directed mission-oriented innovation policies, which themselves are not new. This latest incarnation stems from earlier generations of missions, most notably the space projects led by US and other Western economies during the post-war years (Kattel and Mazzucato, 2018). The recent re-emergence of mission policies reflects a recognition that over several decades innovation policies have been focused on supply driven policy – e.g. via state investment in STI – towards narrow goals of economic growth, and without sufficient focus on diffusion or wider demand side policy in complex, holistic socio-technical systems (Brown and Mason, 2014; Weber and Truffer, 2017). Hence this new generation offers a means of guiding and directing diverse state development in more purposeful and strategic ways (Janssen et al., 2020; Wittmann et al., 2021; Larrue, 2021; Wanzenböck et al., 2020).

Whilst the aim of missions is to move beyond linear STI interventions and towards addressing more purposeful goals, critics suggest a mixed picture as to whether they are succeeding in this endeavour. There are several factors that contribute to this including that mission-oriented approaches remain dominated by STI actors and measures and lack nuance to help policy makers deploy the full range of policy instruments needed to achieve directionality (Bergek et al., 2023; Brown, 2020; Boon and Edler, 2018). Also, idealistic tendencies amongst those STI researchers and policy makers about how new policies come to be implemented in practice (Flanagan and Uyarra, 2016). New innovation frames don't simply displace earlier ones, for example, instead they sit alongside creating points of tension but also, potentially, a degree of risk-averseness in their practical application (Schot and Steinmueller, 2018). Also, context-sensitive institutional changes and decisions taken by actors on the ground have a powerful influence on outcomes.

There is a diversity to mission-oriented innovation policies that is important to note. Several studies have distinguished types of mission policy. Wittmann et al. (2020), for example, distinguish between "accelerator" missions which do remain predominantly scientific and technological in focus, and more "transformational" missions focused on wider systemic change. The more tightly focused "accelerator" type has remained attractive to STI policy makers for their time-bounded and measurable goals attached to targeted scientific or technological investment. Whereas the problems that 'transformer' missions are addressing; the challenges brought about by aging societies or the technological and social challenges around AI, for example, make them more complex and open-ended. As such they are deemed to demand different capacities, including a wider set of instruments or 'policy mixes' that interact in different ways (Russo and Pavone, 2021; Borrás, 2009; Flanagan et al., 2011), and more elaborate forms of governance and coordination (Polt et al., 2019; Wittmann et al., 2020).

The OECD has undertaken the most extensive study of how countries are implementing mission policies. It identifies four types: i) overarching mission-oriented strategic frameworks, ii) challenge-based programmes,

iii) ecosystem-based programmes and iv) mission-oriented thematic programmes (Larrue, 2019, 2021). These types can be seen as further developments of the typology developed by Wittmann et al. (above): for instance, the 'overarching mission-oriented strategic frameworks' are more suited to 'transformer' missions and come closest to the ideal mission approach (Wittman et al., 2020); meanwhile, 'challenge-based programmes' tend to focus on 'accelerator' mission types, though it acknowledges there may be a more nuanced association between different types and policies (Larrue, 2021b).

Whilst challenge-based programmes tend to be more focused, they are also seen to face the most significant problems, including issues with scaling initial experimentation to achieve wider diffusion, which relies on capacities with the programme leadership and coordination to capture and reflect learning to build into new initiatives (Larrue, 2021b). The OECD also highlights how sensitive the more transformer mission-oriented innovation policy is to the specificities of national settings, reflecting a criticism that these policies are often not contextualised enough (Brown, 2020; Edler and Fagerberg, 2017). An aspect of this is institutional in that the design of institutions, their specificity and organisation, influences their effectiveness (Breznitz et al., 2018). In the case of more transformer missions, therefore, the history and remit of an organisation, the goals it adopts, and the activities it pursues all matter in the effectiveness of these approaches (Borrás and Edquist, 2013).

This broad literature highlights several common characteristics and design principles for missions and other transformative frame policies, to include:

- Clear directionality for engaging with the dynamic nature of societal problems or 'grand challenges' and/or moving towards more inclusive and sustainable growth (Haddad et al., 2022; Schot and Steinmueller, 2018; Mazzucato, 2018b; Diercks et al., 2019; Foray, 2018;)
- Multi-faceted policy intervention that acknowledges the need for supply-side and demand-side policies and involving identifying and implementing new instruments as well as measures the phasing out of the 'old' practices and technologies (Haddad et al., 2022; Diercks et al., 2019; Foray, 2018; Mazzucato, 2018a; Kivimaa and Kern, 2016).
- Multi-actor involvement and networks to address a wider agenda for change (Haddad et al., 2022; Kuhlmann and Rip, 2018; Boon and Edler, 2018; Diercks et al., 2019; Grillitsch et al., 2019).

- Multi-level coordination and governance involving 'horizontal' coordinating across policy domains (Kuhlmann and Rip, 2018) and 'vertical' at different geographical scales (Brown, 2020; Borrás and Edler, 2020).

These characteristics essentially describe the tasks and challenge facing STI policy actors who are often chiefly responsible for designing and implementing mission policies, establishing governance frameworks for aligning a broader set of initiatives and actors beyond their usual domain (Boon and Edler, 2018; Kuhlmann and Rip, 2018). In their study about the implementation of the climate goal in the Swedish process industry, Bergek et al. (2023) highlight eight directionality challenges faced in translating more transformative goals into concrete actions. This includes the challenges of handling goal conflicts amongst different stakeholders, identifying realistic pathways, formulating appropriate strategies and measures, and mobilising relevant policy domains and target groups. This research underlines some of the challenges that policy makers face, and that translating missions into concreate actions is highly context-dependent on the capacities and capabilities of actors involved, and their division of labour.

More generally, however, whilst there is a growing theoretical and empirical literature on mission characteristics and implementation, there has been little focus on the individuals and organisations leading these approaches and the capacities and capabilities that come into play when deploying missions (Janssen et al., 2020; Larrue, 2021). There are some notable exceptions including Diercks' (2019) study of the challenges faced in the uptake of systems innovation thinking in the OECD STI[3] Directorate (Diercks, 2019), and the national-level study by Ulmanen et al. (2022) looking at the translation of transformative innovation thinking into policy practice at Vinnova, Sweden's Innovation Agency. These studies emphasis the tensions and challenges that arise as new innovation frames emerge within an organisation. They also highlight the importance of context, and how organisations draw on or extend their capacities and capabilities, to the success of translating transformative policies on the ground.

What then remains a gap – both theoretically and empirically – is how organisations are adapting to deliver these more conscious and complex approaches in real time, and what policy capacities and capabilities they develop for their design and implementation (Kattel and Mazzucato, 2018).

2.2. Conceptualising policy capacities for mission-oriented innovation policy

Policy capacities are perceived as playing a pivotal role in effective governance and seen as fundamental to unlocking innovation (Gieske et al., 2016). As such, policy capacity has emerged as a major concern of governments seeking to address complex problems (Wu et al., 2018; Peters, 2018). The concept of policy capacities draws from public policy and public administration research and most define it from the perspective of governments' ability to *"marshal necessary resources to make intelligent collective choices and set strategic direction for the allocation of scarce resources to public ends"* (Painter and Pierre, 2005 p.2). However, a more expansive definition, highlighting additional factors including government's role in *implementing* policy, as well as making choices, is important when looking at the capacities needed across the complete innovation policy cycle (Wu et al., 2018).

In the innovation policy literature, policy capacity discussion has focused on the role of individual agencies. Thus, for instance, older industrial policy-based discussions focused on the so-called 'embedded autonomy' of central agencies (Evans, 1995). These omnipotent, bureaucratic organisations with their strong links to wider society have been understood as key to the transformative capacity of the state to initiate structural change in the economy (Weiss, 1998). In more recent and STI policy-focused discussions, such notions of capacity have been complemented by what can be called the Schumpeterian alternative. Breznitz has shown that some of the key innovation agencies in the US, Finland, Sweden, Israel, Ireland and Singapore were not central agencies but rather (at least initially) peripheral agencies (Breznitz and Ornston, 2013; Breznitz et al., 2018). These agencies were crucial sources of policy innovations necessary for promoting rapid innovation-based competition through explorations in innovation policy, driven partially by continuous, radical experimentation in their core mission and by the existence of sufficient managerial capacities (or slack) (Karo and Kattel, 2018). Importantly, such organisations developed strong "mission mystique" (Goodsell, 2011), a sense and practice of organisational identity and purpose that allowed them to build capacities to deliver experimental policies.

A further important discussion in innovation policy capacity has focused on the impact of managerial reforms in the 1990s – so-called new public management (Pollitt, 2007; Drechsler, 2005) – leading to the rise of relatively autonomous STI agencies focused on competitive grants, project management skills, outsourcing of some of the key functions (e.g., peer-review) and the rise of market failure-based evaluation frameworks (Kattel et al., 2019; Suurna and Kattel, 2010).

According to the key characteristics of mission-oriented innovation policy described in the previous section, such new policies assume a more expansive definition of policy capacities that focus attention on a

[3] Science, Technology and Innovation.

meta-level of governance and the capacities of a constellation of actors – across public, private and civil society organisations (Kuhlman and Rip, 2018). As such, they can be said to entail the effective coming together of both the endogenous capacities of the organisation(s) leading missions and the exogenous capabilities of a wider network of agents and institutions that contribute to the development and diffusion of innovation and ultimate success of a societal mission (Hekkert et al., 2020).

Thus, policy capacity in the context of mission-oriented innovation policy can be understood as a multi-level construct to include individual, institutional and network capacities and capabilities (Gieske et al., 2016). At individual and institutional levels this includes the use of sense-making for conceptual understanding and to navigate the system or systems the mission is seeking to influence (Edler et al., 2021. Kuhlmann and Rip, 2018). In a wider organisational context lead organisations are also required to support networked systems of governance, thereby exercising their 'transformative power' though establishing frameworks for coordinating complex mission formulation and implementation (Weiss, 1998; Kuhlmann and Rip, 2018; Wanzenböck et al., 2020; Könnölä et al., 2021). Operational capacity is also needed at all stages of mission formulation and implementation including expert functions for evidence gathering, coordination and learning. The timescale involved in missions and the often iterative and dynamic nature of the mission itself means this capacity needs to be agile (with flexibility of systems, governance, and funding) to change direction when needed (Weber et al., 2021). This also highlights the need for ongoing monitoring and evaluation that builds in reflexivity and learning throughout the mission design and implementation (Molas Gallart et al., 2021).

Whilst some of these capacities will already exist in a lead STI institution and in the wider configurations of policy actors, we have very little empirical understanding and evidence of how a lead organisation develops its capacities and capabilities in designing and implementing innovation policy measures of this kind. For instance, it may require an organisation to bring in external expertise, work with new stakeholders, or develop new ways to evaluate policy impact. In turn, these processes of introducing new policy frames, tasks or actors have the potential to be a source of conflict between stakeholders. However, there are few studies that explore the dynamics involved as civil servants and other actors execute tasks to facilitate societal transitions, Braams et al. (2023) being a recent exception. Equally importantly, there is hardly any empirical discussion of the interactions between lead agency and other actors, including other public sector actors. Hence, a further aspect to institutional policy capacities is the degree of ambidexterity, including the ability to innovate and integrate new capabilities with existing capacities (Gieske et al., 2016; Karo and Kattel, 2018). These dynamic capabilities, conceptualised as managerial and organisational routines, are best understood in the wider context of state capacities for existing routines to design and implement policies (Kattel, 2022).

Bringing together the literature on the types and characteristics mission-oriented innovation policies along with literature on policy capacities helps to envisage the sort of capacities and capabilities we might expect to be present or being developed in a lead organisation for such new policies (summarised in Table 1):

- *Navigation and Dynamic Portfolio Management*. Missions are defined by their directionality and goal-oriented nature (Mazzucato, 2018a; Larrue, 2019; Haddad et al., 2022). However, that direction is not fixed; it requires ongoing navigation, sense-making and appropriate frameworks for mission formulation and coordination (Wanzenböck et al., 2020). Navigation relies on capacities and capabilities for deep conceptual understanding of the mission and the analytical skills to identify system conditions, monitor change and influence narratives and direction (Edler et al., 2021. Kuhlmann and Rip, 2018. Kattel, 2022). In this sense, it is closer to the capacity to use a compass effectively than the ability to read a static map. This then includes curation of portfolios of projects utilising and influencing a range of policy instruments and mixes including those in STI but also beyond into other domains (Kattel and Mazzucato, 2018). It also relies on the agility and levers to change direction via governance arrangements and/or within a portfolio (via instrument/investment adjustments) as needed. Missions demand a degree of ambidexterity, i.e. the ability to balance mission objectives, steering mechanisms and incentive structures (Wittmann et al., 2024). Also to balance incremental innovation alongside more explorative actions and activities, e.g. risk taking and experimentation, thereby deploying resources in ways that mean more transformative innovation is achieved (Gieske et al., 2016; Jansen et al., 2005).

Table 1
Framework for analysing policy capacities for missions.

Capacity	Indicators/Keywords
Navigation & Dynamic Portfolio Management	o *Goal-oriented* – in mission and behaviours o *Frameworks* - for mission formulation, system change pathways and lifecycles. o *Portfolio and policy mixes* – combining different instruments including standard STI and beyond (supply and demand/wider diffusion) to create portfolios of investments that address the mission. o *Levers for changing direction* via governance and within a portfolio (e.g. to discontinue investment)
Connecting & Coordination	o *Internal to mission lead organisation* – a shift from organisation as 'machine' to 'living system' (e.g. agile teams, matrix management, end to end accountability) o External Networks (inc. at different scales – local, regional, national) – building legitimacy and accountability for the challenge
Learning & reflexivity	o Ongoing and reflexive learning – performance management, evaluation and learning at project, programme & mission levels o *Influencing/ changing practice* – applying double loop learning to influence internal practice and in government and/or wider network

- *Connecting and Coordinating*. Missions need to be understood in a wider context to include the range of actors that contribute to its formulation and implementation (Wanzenböck et al., 2020). This relies on capabilities in the lead organisation to facilitate both intra- and inter-organisational collaboration (Gieske et al., 2016). *Internally*, this means information sharing and establishing trust by connecting the necessary capabilities (cross discipline, sector, specialist, organisational) needed for knowledge diffusion and innovation (Lin, 2007). Missions also rely on the capacities of a lead organisation to coordinate and nurture numerous *external* collaborative networks including at different scales – local, regional and national (Borrás and Edler, 2020). These informal and formal interactions are key to implementation and building legitimacy and accountability for the mission (Wanzenböck et al., 2020)
- *Learning and Reflexivity*. Mission policies require significant capacities for learning and reflexivity (Larrue, 2021). This includes more formal performance management practices, monitoring and evaluation at programme and mission levels, which in turn informs mission direction and system-change pathways (Kattel and Mazzucato, 2018). It also requires the capacity to stimulate creativity and innovation around the mission process, embedding a double-loop learning approach whereby learning and reflection is being captured throughout, engaged with critically and incorporated into future practice (Jeppesen and Hauan, 2017).

We can anticipate these three sets of capacities and capabilities develop idiosyncratic forms depending on the type of mission-oriented innovation policies (accelerator vs transformer policies) and contextual factors (including division of labour amongst key STI and wider actors).

3. The research case and method

The study explores how UK Research and Innovation (UKRI) developed its policy capacities for delivering the Industrial Strategy Challenge Fund (ISCF). UKRI is a non-departmental public body, created in 2017 by the Higher Education and Research Act, which brought together nine separately chartered organisations (see Fig. 1) – seven individual Research Councils, Research England, and Innovate UK. It had an annual budget of £7908 m in 2021/22[4] and 7463 staff.[5] In addition to the individual entities, UKRI also established a 'central' resource providing strategic and coordination capacity, as recommended in the 'Nurse Review' that led to UKRI's establishment (BEIS, 2015).

The ISCF was the UK's first large-scale Challenge-led innovation programme and initially conceived as an "industry-led" programme to support the government's aim to raise long-term productivity. Established by BEIS and launched in 2016 using funding from the National Productivity Investment Fund (NPIF), the ISCF had government funding of £2.6bn from 2017/18–2024/25. The timeline at Fig. 1 sets out the key dates of relevance to this study. ISCF aimed to tackle four "Grand Challenges" published in the 2017 Industrial Strategy (see Table 2 below), supported via three funding "waves", and with each challenge having several specific goals (e.g., a goal for Audience of the Future was: UK creates 10 % of global creative immersive content). Both BEIS and HM Treasury played a role in scrutinising and approving spend from the Fund, based on prepared business cases. Changes in political climate saw a switch in emphasis from the 2017 Industrial Strategy to the 2021 Innovation Strategy, but did not seem to alter the course of ISCF, though they may have implications at a conceptual level (i.e., for the selection of particular 'Challenges') and in practical terms (the scale of funding or the inclusion of other criteria such as "levelling up").

The case study method has several advantages for this research. It enables the exploration of structures and events in spatial and temporal contexts and provide a rich understanding of how social phenomena are being interpreted and implemented in practice. This case is interesting due to the dynamic setting in which the ISCF was introduced. Whilst Innovate UK was the lead for the ISCF within UKRI, on the ground strategic decisions and the delivery of funding calls was shaped and run by staff from across the different Councils as well as a growing core team at the centre of the new organisation. For this reason, the case study focuses on UKRI, as the lead agency as opposed to a narrower focus on Innovate UK. Also, whilst the institutional setting is UKRI, the study recognises that these policies extend beyond the lead agency, which is reflected in consideration of how UKRI is evolving its capacities to deliver across a wider set of domains.

An abductive approach enabled a combination of inductive and deductive reasoning in observing UKRI's interpretation of missions in a dynamic policy setting where political shifts and organisational changes contribute to the evolution of strategic frameworks for innovation, and therefore missions sit amongst more traditional frames of innovation and modes of delivery. The study also looked for evidence of how UKRI was evolving its practice for delivering missions against the sorts of capacities identified in the theoretical analysis of missions and policy capacities, set out in the previous section.

The empirical evidence reported draws upon documentary review, analysis of specific mission/Challenge approaches to supporting innovation (e.g. types of funding instrument and engagements approaches), and interviews with senior UKRI staff (14). Interviews were undertaken by the first author and conducted and recorded via zoom. They involved senior figures who were involved in the programme: three Industrial Strategy Challenge Directors, senior managers[6] with responsibility for innovation in Innovate UK, and senior managers with responsibility for innovation in three of the Research Councils and BEIS. The choice of interviewees was initially based on identifying those individuals with lead responsibility in challenge-based innovation; further interviewees were added via a snow-balling sampling approach by asking for suggestions from interviewees based on subject matter and areas of focus (Weiss, 1994). This was deemed an appropriate method in this context where representativeness was not a necessary standard for the overall sample (Small, 2009). Interview transcripts were initially analysed against capacities identified in the theoretical analysis, using the indicators and key words in Table 1 as a coding framework. The interviews were then coded a second time using open coding to allow new themes and connections between themes to emerge.

4. Developing policy capacities for mission-oriented innovation policy in UKRI – Case Findings

Public policy evolves over time, interacting with political changes, budget cycles, institutional life cycles and other dynamics (Flanagan and Uyarra, 2016). New innovation policies, including missions, are introduced and adopted in pre-existing institutional frameworks that will have also evolved over time, and alongside existing policies and instruments, each with their own logics, and some of which might be in tension with new policy frames (Uyarra, 2010). Therefore, how an organisation is developing capacities and capabilities for challenge-led missions needs to be explored and understood within this dynamic context. Following this logic, this section starts by examining the changing institutional context of UKRI before setting out the different interpretations of the role of ISCF that were evident in the interviews, and how that evolved over time. It then goes on to explore whether the capacities and capacities for challenge-led missions were apparent, or not, in this case.

4.1. UKRI and the bringing together of research and innovation

The creation of UKRI marked a new phase of collaboration amongst the research councils, Innovate UK and Research England, and against a backdrop of change as the UK exited the EU and, at the time, its future participation in EU Horizon funding was uncertain. Whilst the nine individual organisations were working together prior to UKRI, its creation marked a shift in identity and function given its broader remit to set direction for a complex research and innovation system. This fusion was significant for the constituent organisations, each with their own core remit, organisational culture and practices. The seven individual Research Councils were originally established as main funding bodies of research activity for the disciplines/fields they support, and this role largely continues. The connection between research activities they support and wider societal change (e.g. via knowledge exchange and collaboration) had always been present, but the traditional justification for this investment was the creation of knowledge as a public good rather than for purpose-driven innovation. Research England's role is primarily to support English higher education providers (HEPs) alongside equivalents in Wales, Scotland and Northern Ireland and as part of that to monitor research (and more recently knowledge exchange) activity and performance and to distribute longer-term formula funding streams. Innovate UK, the UK's Innovation Agency, was originally established as the Technology Strategy Board to support innovative technologies, and today remains primarily focused on driving economic growth through support for businesses. This is relevant in thinking about how UKRI conceptualises its roles in research and innovation,

[4] For 2021/22. This is £403 m lower (5 %) than 20/21 budget. Primary accounted for by a £284 m reduction in Overseas Development Aid (ODA) (BEIS, 2021c).

[5] Figure for 2019/20 (UKRI, 2020).

[6] UKRI staff interviewed were predominantly Director level (Senior civil service equivalent) or Associate Director.

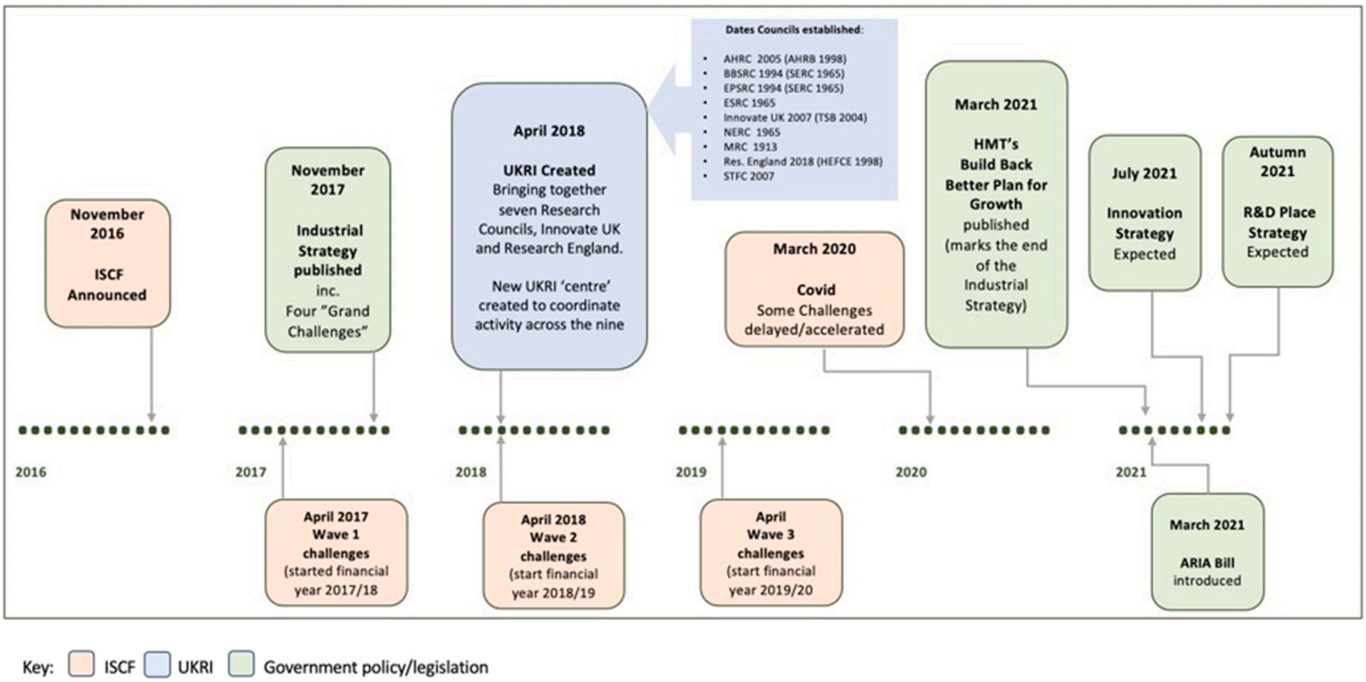

Fig. 1. Timeline showing the creation of ISCF and UKRI in the context of Government policy for Industrial/Innovation Strategy. (Adapted from a version in the 2021 audit of the ISCF (NAO, 2021).)

Table 2
ISCF Challenges by Funding Wave.[a]

Grand Challenge	Wave 1	Wave 2	Wave 3
Aging Society (£607 m)	Medicine Manufacturing (£207 m)	Early diagnosis & precision medicine (£223 m) Healthy Aging (£98 m)	Accelerating Detection of Disease (£79 m)
AI and data (£302 m)	Robotics and AI in Extreme Environments (£112 m)	Audience of the future (£39 m) Next generation services (£20 m) Quantum Technologies (£20 m)	Digital Security by Design (£70 m) Commercialising Quantum Technology (£153 m)
Future mobility (£770 m)	Faraday Battery Challenge (£318 m) Self-driving Vehicles (£26 m) National Satellite Test Facility (109 m)*		Driving the Electric Revolution (£80 m) Future Flight (£125 m)
Clean Growth (£1049 m)	Self-driving Vehicles (£26 m)	Transforming construction (173 m) Transforming food production (£90 m) Prospering from the Energy Revolution (£108 m)	Industrial Decarbonisation (£170 m) Smart Sustainable Plastic Packaging (£60 m) Low Cost Nuclear (£235 m) Manufacturing Made Smarter (£147 m) Transforming Foundation Industries (£66 m)

*Programmes not established through a challenge-led approach.
[a] Based on UKRI sources and the National Audit Office Report on ISCF (NAO, 2021).

particularly because none of the organisations were originally set up with an expansive role in innovation policy and its impact on society, Innovate UK coming closest. Perhaps reflecting these different origins and communities, UKRI's establishment was not without controversy. One key element of this was the merits or otherwise of including Innovate UK in UKRI along with a series of concerns about the autonomy and strategic flexibility of individual Councils/entities which was retained in the legislation that established UKRI. For some, these concerns led to compromises in the management and governance arrangements which notably includes an Executive Board comprising the UKRI Chief Executive Officer as Chair, the Executive Chairs of each of the nine organisations along with senior officers from BEIS (Wilsdon, 2016). In addition, a Governing Board made up of the Chair, CEO, Chief Finance Officer and between 9 and 12 independent members drawn from higher education, industry and commerce, policy, and charities and other non-governmental organisations (BEIS, 2018).

The interviews with senior staff across the organisation indicated that UKRI was still evolving its role in the wider innovation ecosystem and the balance across the core aspects of its remit: namely science (or research) and innovation. A small number of interviewees observed that research and innovation appeared to sit a little uneasily together in UKRI. The 'how' of innovation investment was explained by several respondents in the dichotomous language of science push (Research Councils) or business pull (Innovate UK), though there was also recognition it was not this simple in practice with all nine organisations investing in and contributing to research and innovation in multiple ways.

There was some evidence of tension in the balance of UKRI investment between science (or research) and innovation and therefore between individual organisations within the overarching structure. This may have been felt keenly due to UKRI's budget being under pressure at the time (in this instance due to changes to Overseas Development Aid having led to a reduction in GCRF), though there was also reference to a perceived imbalance of power on the UKRI Executive, which was seen by some to favour investment in research over its role in innovation, if only by voting numbers (i.e. seven Research Councils to one Innovate UK), leading one respondent to remark that "UKRI management don't see innovation as a core part of their thing" (Interviewee 10).[7] This was relevant because ISCF marked a new approach to supporting innovation and required different ways of working to bring UKRI's key capacities in research and innovation closer together. Those involved in programme direction felt this tension in the balance of the two quite keenly, and as one respondent put it: "This integration of innovation and research is going to be absolutely critical and I'm not sure UKRI is really … well, they can continue to learn about how to do that." (Interviewee 2).

A further tension in the UKRI structure at the time, was a lack of clarify about the role of a growing team of officers at the centre of UKRI. Several respondents remarked on the size of the team with the intimation that it was too big, but more importantly perhaps, it had not yet fully established the sort of agile and/or matrix ways of working across the nine entities that some felt was required to support ISCF and the new ways of working it demanded: "we have had no input from UKRI central, and it would seem that is one of the problems of framing-up that mission/challenge based stuff" (Interviewee 1) and also the recommendation that "[the] first thing it's [UKRI] got to do work out its own devolved processes." (Interviewee 9).

4.2. Balancing agendas at ISCF Programme level

From the outset ISCF was an ambitious programme with significant public investment and oversight. Backed by £2.6 billion of public money, the UK government was looking for ISCF to contribute to an increasing number of objectives, including its commitment to raise spend on R&D to 2.4 % GDP alongside commitments to net zero and 'levelling up' agendas (NAO, 2021). However, here it was also recognised as a potential point of friction with the concept of directional, long term 'missions' and systemic change and raised potential operational challenges for more dynamic portfolio management, in that private sector interests needed to be considered around decisions about discontinuing underperforming projects investments:

> [S]ome of the stuff we have been doing are sort of missions and challenges, but how do you develop a programme which is predominately co-funded? I think what we have been trying to do with the challenge fund, you know, one of the requirements, from both politicians but also officials was how do you help encourage industry to invest?.
>
> (Interviewee 5)

> [A] big drive to make [ISCF] happen was getting industry money on board. That driver to make it happen was, 'can you prove to me for every £100m of government investment you've got £200m from X, Y or Z'. That was the driver, not a whole system approach.
>
> (Interviewee 9)

In reality, several respondents acknowledged that ISCF would seek to address multiple agendas, including changing political priorities in the context of Brexit as well as contributing to societal challenges of varying shapes and sizes, but also saw the complexity in operationalising that through managing a challenge or mission portfolio:

> [B]ecause there [are] so many good things one can do, what you're trying to do is think about 'how does this deliver against multiple agendas?' [W]hat does that mean in the context of levelling up? What does that mean in the context of carbon reduction? What does it mean in the context of skills? And when we were trying to make choices, we were trying to look at those things and so part of the portfolio is about – does it address several things?.
>
> (Interviewee 5)

The overarching driver to unlock investment from the private sector was not so surprising given the state's role in innovation has become more focused on finding ways to incentivise the private sector to invest, and indeed some see this as a factor in the popularity of mission approaches (Edler et al., 2021). However, the expectations to meet different policy agendas in ways that are synergistic creates potential for tension between goals and conflict between stakeholders around priority setting (Bergek et al., 2023) which would require capabilities for careful management.

4.3. Evolving narratives around ISCF and challenge types

Evolving policy narratives and approaches for Challenge identification was a further contextual factor influencing the development of capacities and capabilities. The pace of ISCF's arrival and its coincidence with UKRI's creation meant a pragmatic formulation of Wave 1 challenges. This first wave was developed before the industrial strategy was launched and therefore lacked a clear framework guiding the types of challenges selected. Instead, individual challenges were developed drawing on organisational knowledge and expertise from Innovate UK, with input from the seven Research Councils. As one respondent put it "for Wave 1 (which began financial year 207/18), … I guess [it was] more, you know, 'what's the big, interesting things that we all sort of know that the UK should map against'" (Interviewee 3). It is perhaps as a result that challenges selected and strategies to address them were framed fairly narrowly in terms of accelerating technological innovation through science and business innovation, rather than a broader set of measures delivering wider societal change (see Table 2).

Wave 2 (financial year 2018/19) then coincided with the launch of the Industrial Strategy in 2017/18 setting out the 'Grand Challenges' against which all challenges have been mapped subsequently (see Table 2 above). It also corresponded with the Industrial Strategy's "Sector Deals"[8] with challenges aligning with the promises set out in those: "…and then Wave 2 was launched at the same time as the Industrial Strategy so had they had a bit of a sector flavour" (Interviewee 5). A wider consultation with stakeholders helped identify Wave 3 Challenges (financial year 2019/20), drawing more than 250 responses which were then sifted down to 10 challenges, including areas that were 'matched up' or synthesised e.g. Foundational Industries, which combined several submissions. A shift can be seen over these three waves, in the deliberative approach taken to select challenges and the challenge 'type' being supported, with implications for the sorts of capacities and capabilities being called for or utilised.

Whilst overall, many challenges retained a focus on either technological solutions or transforming sectors, akin to the "accelerator" type of Challenge or mission referenced in the literature (Wittmann et al., 2020), a small number were broader and framed with a more societal change or "transformational" emphasis. An example is Healthy Aging which, due to the nature and breadth of the challenge, incorporated more partners and a wider conceptualisation of innovation and enterprise. This said, from the examples looked at in more detail, even those described in narrower technological language were not solely focused on technological solutions, and the interview responses indicated an appreciation that the pathways or possible solutions to addressing all challenges were multiple and more complex:

[7] To maintain anonymity, interviewees have been randomised and allocated a number. Where a quote may indicate the interviewee by its content they have been labelled 'Interviewee Anon' to ensure other quotes by the same interviewee remain anonymised.

[8] Sector Deals are a partnership between government and industry to focus attention and resources on sector-specific issues to boost productivity and innovation. Relevant Challenges in ISCF have sometimes been aligned to or an integral part of a Sector Deal (e.g. Construction Sector Deal and Transforming Construction Challenge)

> [T]rying to put all of that sort of stuff together in a holistic, whole system way where whole system doesn't just mean the technical system and it means you know what the consumers actually want to pay for, what finance want to finance, what regulators want to regulate, and [then] how does the technological piece play into that.
>
> (Interviewee 2)

The evolution of the programme, as it gradually encompassed a more diverse set of mission or challenge types over successive waves, was echoed in the discourse around the programme itself which at the time of the interviews had shifted to being about addressing *societal* challenges. For those working at the heart of UKRI and preparing for how ISCF would feature in the forthcoming Innovation Strategy (again, at the time of the interviews), this shift from technical challenges to sectors and then to societal challenges was recognised. ISCF at this stage is understood as an "experimental way of thinking about innovation, so rather than targeting sectors or disciplines, as we were more used to in the UK, we targeted Industrial and societal challenges or problems."

(Interviewee 3)

4.4. Policy capacities for mission-oriented innovation

This section provides an analysis of the findings on policy capacities for missions, mapped against three sets of capacities identified earlier (set out in Table 1). The more detailed findings are summarised in Table 3 below, including positive indications of where policy capacities were present or emerging alongside identified gaps or where they were less prevalent.

4.4.1. Policy capacity I: navigation and dynamic portfolio management

4.4.1.1. Goal-oriented/innovation frames. Mission-oriented innovation policies are not delivered in a vacuum, they emerge and are embedded in institutions that have norms, belief systems and practices, all of which influence how investment in innovation is perceived (Larrue, 2021). As the previous section illustrates, ISCF wasn't wholly viewed in UKRI in terms of addressing a societal challenge or effecting systems change. In fact, several respondents described UKRI's role in innovation and the purpose of ISCF primarily in terms of driving economic growth and "market fixing". As one respondent made clear "economic outcomes are a massive thing for UKRI" (Interviewee 3). However, there was nuance in this, including reference to "sustainable economic growth" being the preferred terminology in strategy and policy documents, though also reference to any ambiguity in this terminology being seen as advantageous. Another that "where there is a market failure, as you know, government will stimulate the market by creating an artificial market" (Interviewee 4). However, those more closely involved in the purpose and strategy for ISCF recognised the complexity of innovation and the role of UKRI shifting to more purpose-driven approaches: "[There's a] real push to think more in terms of systems… [an] appreciation of the complexity of innovation and …[for] innovation with purpose" (Interviewee 3). This more transformative language was also echoed in how most respondents described the role and capacity of UKRI as "investors" in research and innovation, not merely "funders" of innovation.

4.4.1.2. Frameworks. ISCF challenges were intended to be long term, even though the immediate funding commitment for them might be just 3–4 years duration. This created difficulties for those facilitating the process in terms of the strategic-level navigation needed for formulating the challenge and carving out a route map, particularly when medium- and longer-term needs were less clear and multiple possible pathways existed. Respondents working at the centre of ISCF reported a more open-ended way of working as something they were moving towards, including seeking to configure more agile principles and practices into

Table 3
Interview coding: summary of findings.

	Evidence of policy capacities for Missions/ Challenges	Areas of weakness/gaps
Navigation & Dynamic Portfolio Management Keywords (from Table 1): - Goal-oriented - Frameworks - Portfolio - Levers for changing direction	- **Goal-oriented** – set of challenges with clear goals. - **Frameworks** – strategic & programme frameworks for Challenges are evolving i.e. incorporating directionality beyond immediate funding cycle & exploring ways to deal with uncertainty/ multiple pathways (though not yet embedded fully in practice). Level of autonomy in decision making at mission level is fuzzy. - **Portfolio/mix** – standard UKRI/STI investments though some examples of Challenges innovating with new funded entities and ways of integrating research and innovation (e.g. short and long loop research investments working alongside demonstrators and other innovation investments. Also seeking to influence other innovation policy levers/levels e.g. regulatory policies, local innovation systems. - **Levers for change** – can be built into funding design e.g. state-gates. Discontinued investments are still a small minority. Within portfolio, agility relies on investment lead university/ PI flexibility to change direction or focus.	- **Goal-oriented –** on balance most challenges are technology or sector focused. More traditional frames of innovation are dominant though some appetite for more transformative mission approach. - **Frameworks** – engagement to inform decisions around Challenges could be wider (and is recognised as a shortcoming or gap). Governance around missions is predominantly government, business and research actors. No guidelines in place to shape engagement at Challenge level (though evidence based on select Challenges suggests it is broader in some cases, though not consistently) - **Portfolio/mix** – Limited to UKRI capacities for investment. Within this - tension in balance of research (long term, "blue sky") and innovation funding in Challenges. Constrained in part by what UKRI funding can do and decision-making rules/culture (though informal workarounds are common). Need for a "playbook" of funding instruments for use across Challenges. - **Levers for change** – at programme level: constrained by limited capacities. Scale of funding means greater HMT scrutiny of changes to original business case. However, COVID-19 has provided a precedent for greater flexibility. Lighter touch process for revision of business case and higher tolerances for budget virement are needed. Other barriers to portfolio change: potential for tension with private co-investment (i.e. barrier to stopping investments that are co-funded)
Connection & Coordination Keywords (from Table 1): - Internal to UKRI - External Networks (inc. national & local/ regional)	Internal - Appointment of Challenge Directors (CD) with external business expertise seen as positive overall. Becoming an expert community in own right within UKRI. - Cross-UKRI engagement in challenges has	Internal - Challenge teams not fully integrated in UKRI which creates unnecessary barriers. - Ability of Challenge Teams to draw on resource across organisation is constrained by UKRI silos and hierarchies and needs more

(continued on next page)

Table 3 (*continued*)

	Evidence of policy capacities for Missions/ Challenges	Areas of weakness/gaps
	increased as challenges have evolved (specific references made to positive of growing involvement of social science input into innovation & diffusion). External networks - Differs at different levels: range of actors narrower at ISCF programme governance level and increases at Challenge portfolio level. - Evidence of involvement of local actors in some challenge investments (i.e. local authorities and expectation this will expand with 'place' emphasis at national policy level) - Challenges connect well with other parts of the innovation infrastructure e.g. Catapults - Examples of novelty in engaging broader stakeholders in challenges (e.g. use of arts, performance) - Regional Managers in Innovate UK and EPSRC (separate teams) connect local actors to the Challenges and enhance UKRI knowledge of local strengths, ambitions etc. (not limited to Challenges).	agile teams/matrix approaches. - Communication across Challenges is ad hoc. External networks - Depends heavily on Challenge Director capabilities and network. Some have Deputy Directors (also industry or internal). May need to broaden expertise to achieve real systems change i.e. pilot Directors or Deputies from other sectors/backgrounds. - Little to no delegation of Challenge responsibilities to actors beyond UKRI. Some delegation and devolution of funding to be dispensed by portfolio investments (not systematised at this stage) - Governance needs to reflect wide range of stakeholders for each Challenge (going beyond usual actors). - Scope to expand engagement of local actors in shaping Challenges. Existing capacities/ capabilities could also play more of a role i.e. Catapults and Regional teams. - Danger of 'missions and place' being overly politicised as a blanket requirement, though Innovation Strategy and Place-based R&D Strategy may clarify.
Learning & reflexivity Keywords (from Table 1) - Ongoing and reflexive learning - Influencing practices	Evolution of Challenge approach - Challenge types have evolved since first wave – including some that are societal and/or more complex. Ongoing & reflexive learning - Performance management becoming embedded in Challenges – logic models, learning frames and evidence of date being used to inform direction. - Example of sharing learning around user-centred Design across Challenges with the aim of embedding that further Influencing change in practice - Active portfolio management – i.e. Challenge Directors sharing monitoring data collected with portfolio investments – leading to changes in focus/new opportunities	Evolution of Challenge approach - Overall emphasis still balanced towards science and technological innovation where UKRI capacities are strongest. Ongoing & reflexive learning - Evaluation of ISCF underway (at time of writing) - Limited evidence of systematic sharing of practice across Challenges - Limited evidence of systematic embedding of learning within each mission Influencing change in practice - Little to no evidence of formal change in standard procedures i.e. decision-making practices to ensure highest quality aligning with Challenge goals (some examples of gradual change in norms i.e. willingness to look at balance of portfolio alongside quality but not consistently applied) - Little evidence of ambidexterity in balancing mainstream policy capacities with more experimentation and risk taking (however, would warrant further research to fully determine this aspect).

the governance and programme architecture. This, however, was a work in progress:

> Big missions: let's call it a 10-year strategic goal. To do that, there are all the steppingstones across that big river that you need, from UKRI and beyond …. the reality with those steppingstones is a colossal amount of management to make sure we're spending on the right things. I think governance in one way is misleading on this because [what] you need to do is think about–- what's the big framework to help these things to happen.
>
> (Interviewee 8)

There was some recognition that the existing strategies and frameworks in place for investing in research and business-led innovation were not sufficient for addressing challenges over the longer-term, and that alternative governance models were needed, but also that what discussions on what those frameworks might look like were early stage at best.

4.4.1.3. Portfolio/mix. The portfolio mixes in challenge areas remained predominantly STI focused, utilising standard UKRI research or innovation investment tools and mechanisms. However, within some of the challenges there was evidence of innovation in how investment in research and innovation was being conceived or becoming more woven together as part of a portfolio management approach. This included new types of investment, or changes in practice within existing investment models that create shorter feedback loops than traditional research investments:

> [Specific named research investment] is a £10 million-ish programme with a very different approach to research. With [Investment Name] we said – 'look, we need to figure out what is known about [Challenge relevant] systems and about how they can scale across a whole system, and we need to know about the technology and the financing and the business models and the consumer engagement, from this whole systems perspective'. This kind of synthesis approach is very interesting where it's bringing the whole system knowledge to the party and trying to make sense of all of that. There are bits of primary research going on as well. So, they're working with the demonstrators to test what's going on in the real world, who can swap out what they're trying or develop new theories of change. [Y]ou don't [just] need a different kind of program but you also need a different kind of leadership. [The] PI is very open to sort of doing things in a different way, to create a flexible program that learns and that is able to flex resource and funding.
>
> (Interviewee 2)

This evolution in the design of funding instruments and calls might have gone further if Challenge Director had more freedom to create their own approaches or access a more extended "playbook" of instruments. However, internally this met with resistance on a couple of different fronts including UKRI efficiency agendas which were aimed at reducing the overall number of schemes and calls. Also, different perspectives were expressed amongst individual research councils and Innovate UK as to the whether the balance of investment in Challenge areas was appropriate, with some questioning whether there needed to more investment in fundamental research to provide the scientific knowledge for future technologies. There were clear politics in play in how the Challenges were being governed within UKRI.

4.4.1.4. Changing direction. ISCF budgets were considerably larger and this fact, along with the ties to the government's Industrial Strategy, meant they were accompanied by a higher degree of scrutiny from both BEIS and HMT. This was the case at both the business case approval stage for Challenges as well as in an ongoing sense. A National Audit Office Report on ISCF highlighted the inconsistency in the government's short timescales for spend and the lengthy process to approve a business case and/or for UKRI to get the funding out of the door (NAO, 2021). However, this rigidity was also a problem once the Challenge was progressing:

> How do you then create enough autonomy within let's call it the mission framework that allows people to go, I expected £40 million to be done on this, but actually something's happened to the world and am I going to divert [spend] over here. How do you do that in a way that's agile and flexible enough that allows people to pivot because the world changes.
>
> (Interviewee 8)

As with many areas of public investment, COVID-19 proved to be a stress-test for the flexibility built around the funding and UKRI was given approval to vire resources during this time. This had the further effect of highlighting the need for higher degrees of funding certainty over longer timescales term for the Challenges, as well as tolerance around spend to enable agility; capacities which are important for implementation of mission policies.

4.4.2. Policy capacity II: coordination

4.4.2.1. Internal coordination. There was evidence to suggest strong engagement across UKRI in the shaping of Challenges. However, difficulties were being experienced at an operational level. As one interviewee put it: "Challenge Directors were hired to do a job and it's certainly the case that the [UKRI] system wasn't set up for them to do it, at all" (Interviewee 10). The 'system' here included internal policy and operational procedures and norms which were perceived as a barrier and not providing the agility needed to work across the organisations of UKRI:

> There are some things [about Challenge Directors] should have helped, that is that they're all used to working in goal-oriented, multidisciplinary, matrix managed or collaborative teams. So, they're actually used to this idea that you talk to some people who work over there in that bit of the organisation, and some people who live over there with that bit the organisation, and they come together to work as a team. Unfortunately, UKRI as an organisation does not recognize that and it has struggled with it.
>
> (Interviewee 10)

On the design of funding instruments, one interviewee described it as about the ability to forge a programme using all the options in the "UKRI tool kit" and therefore highlighted the need for UKRI to create a "playbook" to enable Challenge teams to navigate the existing funding instruments and approaches available to them.

4.4.2.2. External networks including 'vertical' and 'horizontal' coordination. As the earlier section of this paper indicated, missions need expanded forms of governance and therefore make different asks on the organisations leading the mission (Kuhlmann and Rip, 2018). In the case of some ISCF Challenges, there was recognition that this meant a larger role for a broader set of external actors than might have traditionally been in case in developing funding calls focused on industry or universities, and with that a more complex set of engagements to manage:

I think in a lot of these missions, certainly the sort of more society facing ones, you know, perhaps not the ones simply trying to develop a particular industrial sector which can be a bit more traditionally formatted, but these big, society-facing ones, you need to work with a whole raft of new types of organisations. You know we're not only working with local authorities and businesses and research, we are also working with lots of NGOs, consumer groups were working with stage companies who are helping with consumer engagement [and] were working with researchers who are right at the sort of cutting edge of delivering this stuff. (Interviewee 2).

Furthermore, 'place' and the relevance of multi-actor, multi-level governance was recognised as significant in several of the challenges, including those that were engaged in local systems change. However, despite being one of the pillars of the 2017 Industrial Strategy, and its growing significance at the time with the levelling-up agenda, place wasn't an explicit feature or driver in the overall governance and implementation of ISCF. The tensions around this were apparent as ISCF operated predominantly at a national level and yet several challenges warranted a more place-based approach with local actors playing an active role:

[There's] an interesting tension between what's coming from the top and what's happening on the ground – a lack of connectivity between the trinity of actors in places: local governments, central government and then the R&D system. [It] needs more localised experimentation because local contextual factors matter. [But] it's unclear how much traction 'missions' have in places partly because of dysfunctional relationships between Whitehall and different parts of UK. [The] system's out of alignment and therefore UKRI levers don't have maximum effect (Interviewee 7).

Place has become more of a feature of research and innovation funding since the interviews took place, however even at that time UKRI had capacities that could support more experimental and contextualised ways of working at local levels – the Connected Place Places Catapult (CPC) for example, that had a system-wide role in supporting place -based innovation agendas.

4.4.3. Policy capacity III: learning and reflexivity

The management and governance of ISCF evolved over its duration, influenced by a range of factors including changes in the wider policy environment, major events/shocks (e.g. COVID-19), along with internal learning (both within UKRI and in the individual challenge areas). At the time this research took place, UKRI had commissioned an evaluation of ISCF, which hadn't yet published its evaluation framework.,[9] though interview sources indicated that the initial evidence was positive, suggesting that "missions work" according to one respondent.

Within challenges, there was evidence that evaluation and performance management approaches were beginning to be used to monitor programme and inform decisions about how the challenge evolved, though this was a relatively recent introduction. This included the use of logic models and theories of change along with ongoing monitoring to inform next steps:

[9] The Evaluation Framework Report, produced by RAND Europe and Frontier Economics has since been published on the UKRI website at https://www.ukri.org/publications/evaluation-of-the-industrial-stategy-challenge-fund-iscf/. As has the ISCF Process Evaluation Report, published in 2023.

> One of the other things that is different about the Challenge model and challenging for, as far as I can see, every bit of UKRI, is that emphasis and the embracing of performance management– in benefit analysis, benefits capture, impact culture. So that's a big deal and, to be honest, I don't think that's been appreciated by the other bits UKRI.
>
> (Interviewee 10)

However, reflecting on what were fairly standard approaches to evaluation that aligned with the HM Treasury's Magenta Book, some interviewees highlighted the difficulties with aligning evaluation and long-term mission approaches:

> I think one of the issues is unrealistic expectations of what an evaluation can show you after 2, 3, 5 years even [in terms of attribution]. I think is that's the bigger issue.
>
> (Interviewee 9)

> Our logic model extends out by 15 years or so, you know, in terms of some of the long-term impacts that we think will generate but also the way we are tracking our KPIs, some of those are just really difficult to do to be honest. You know if you make a change locally, how much of that change can you attribute to the local impact [verses] what's going on nationally? …so, it's a really complex problem to evaluate.
>
> (Interviewee 2)

Within UKRI itself, embedding learning and influencing change in operating systems and routines was reported to be a work in progress. One example was the balance between goal-oriented investment and current operational systems for programmes design and decision-making. Challenge Directors were empowered to take decisions around the overall shape and direction of the challenge. However, funding protocols meant they didn't always have a say in final decisions about what got funded, even though they were ultimately charged with delivering the goal/challenge and making sure the individual projects cohere to the common purpose. This was primarily about culture and norms at organisational and wider research sector levels, though it can get muddled in issues of interpretation around the Haldane Principle and academic freedom, which remain firmly rooted in decision-making practice.[10] In reality, UKRI has the capacity to evolve how it makes funding decisions for mission-oriented innovation but knew it would need to take the wider research and innovation community with it in any changes introduced.

5. Discussion and conclusions

This article has explored how UKRI, the organisation leading mission-oriented innovation policy approaches in the UK, developed its policy capacities and capabilities for such new initiatives whilst delivering the Industrial Strategy Challenge Fund (ISCF).

Agencies like UKRI are seen as important factors of success in the development, coordination, and delivery of national innovation policies; "as 'change agents' they bundle expertise, orchestrate innovation processes and serve as liaisons across sectors and levels of activity" (Breitinger et al., 2021, p.7). However, missions present a challenge to the structure, culture and remits of those leading on STI policy and implementation and demand a shift in approach. Fundamental to this is moving from investing mainly in supply-led or 'frontier' innovation to a more holistic approach, balancing supply and demand-led policies (Boon and Edler, 2018). A key part of achieving this balance is involving multiple actors across different policy domains to design and address the mission (Wanzenböck et al., 2020). What follows are some conclusions on the specific case study followed by reflections on the literature and areas for future research.

5.1. Research question: UKRI case study

Our main research question focused on what kind of capacities and capabilities are science and innovation funders like UKRI developing and utilising in implementing 'mission' approaches, and what challenges they face? With mission-oriented innovation policies, context matters. The introduction of the UK's Industrial Strategy Challenge Fund coincided with the creation of UKRI, a new entity combining a set or organisations with individual remits - Research Councils focused primarily on research investment, Innovate UK on business-led innovation, and Research England supporting the sustainability of the sector through block grants to universities in England - and each retaining some individual autonomy in the internal governance structure. This evolving organisational context has impacted on the overall effectiveness of Challenge approaches, similarly as Borrás and Edquist (2013) suggest, influencing the framing of Challenges and creating a degree of resistance in the creation of new working practices and integrated mechanisms for research and innovation.

In terms of the mission typology, the ISCF has included mainly "accelerator" type missions (Wittmann et al., 2020). This said, since its launch, wider policy changes including a shift from an industrial to innovation strategy in the UK, as well as incremental learning from within the programme itself, has begun to influence the framing and narratives around ISCF towards more 'societal' and 'transformative' language focused on wider socio-technical system change. Some of the challenges in the ISCF were more "transformer" types (e.g. Healthy Aging with is broader conceptualisation and strong emphasis on collaboration), which require more complex governance and coordination (Wittmann et al., 2020) and therefore, potentially the ability to tailor and draw on different capacities. However, whether this challenge ultimately achieved this was beyond the scope of this particular study and would require more in-depth research of individual challenges.

We proposed that the policy capacities required for missions include navigation capacities, the capacity for coordination – internally, across external networks – and capacities to embed learning and reflexivity. Based on interviews, we can conclude that the range of policy capacities needed for missions were not present in UKRI. There was awareness that these capacities were needed and some minor evidence of their emergence, particularly via the actions of the Challenge Directors as they test the boundaries of what is possible to deliver through UKRI. However, there is also clear evidence of tensions between individual actors and organisations which creates barriers to change. Looking at the categories of policy capacity in turn, we found that:

- *Navigation/Portfolio Management*: UKRI's capacities for formulating and implementing large-scale complex investment are developing, though with some resistance based on established ways of doing things and a lack of autonomy. Examples given from some of the challenges suggest Challenge Directors influenced the ability to innovate and create new ways of engaging partners, including devolving funding via larger investments. However, more generally, its ability to deliver at pace and with agility is constrained by its lack of capacities and constraints caused by other actors, notably in BEIS and HM Treasury. This includes long timescales around business case approvals and micromanagement of delivery, taking away from the ability to experiment and agility needed in challenge implementation.
- *Coordination:* coordination in delivering ISCF Challenges depended in significant part on the framing of the individual challenges and each Challenge Director's capabilities and network. Challenge Directors, brought in from industry for their systems knowledge and

[10] The Higher Education and Research Act 2017 that established UKRI also embedded the Haldane principle which indicates that "decisions on individual proposals are best taken following an evaluation of the research quality and likely impact [..] such as a peer review process" (HERA, 2017: Section 103 (3))

networks, have created positive momentum to accelerator and sector Challenges, and potential solutions. However, as the ISCF shifts to include more societal Challenges, UKRI may want to balance their industry expertise with other forms of system knowledge for more transformational challenges. Whilst their arrival is proving to be a disruptive force within UKRI, similar to positive role played by peripheral agencies described by Breznitz discussed above, they also experience resistance as they unsettle established power bases, cultures and practices. This resistance emerges from public administrators finding it difficult to legitimize the tasks Challenge Directors look to introduce in addressing their Challenge (Braams et al., 2021).
- *Learning and Reflection:* UKRI is beginning to develop more of an evaluation and learning culture and there does seem to be some read across from this to how ISCF is operating (and/or vice versa). Whilst evaluation methods are being utilised, one interviewee suggested this hadn't been from the outset of ISCF but rather a more recent introduction. Also, it was evident that the approach to evaluation was based on standard summative evaluation approaches rather than the more formative models that are becoming associated with transformative mission policies (Molas Gallart et al., 2021; Rohracher et al., 2023)

UKRI was established in parallel with the introduction of the ISCF and perhaps has been doubly challenged on establishing the sorts of capacities and capabilities required at a time of significant change. That said, the creation of UKRI was also an opportunity to better integrate research and innovation investment and create a more agile central core to support matrix working across the nine organisations and beyond, as recommended by the 'Nurse Review' (BEIS, 2015).

Missions have tended to prioritise STI based approaches to innovation, though this is beginning the change. The potential of missions lies in their ability to enable more integrated innovation that brings together supply-led and demand-led or wider diffusion of innovation, thereby incorporating a holistic set of assets and intelligence in addressing challenges. Boon and Edler (2018) argue that demand should be at the core of challenge-oriented policy. Whilst those interviewed in UKRI understood the need for a wider lens on the challenges to inform action, this was rarely articulated in terms of a need to focus on what might be considered demand conditions or activities. However, a couple of interviewees did recognize the potential of a more diffusion-oriented approach to innovation that would incorporate more experimental frameworks involving different knowledge combinations at different scales (Brown, 2020; Brown and Mason, 2014).

This potential had begun to be demonstrated in those challenges exploring place-based investment using key investments like Living Labs and Clusters and could also be bolstered by better integration with the wider UKRI infrastructure including Catapults and other place-based investments introduced in alignment to the Levelling-up agenda, for example. However, this is only possible if the organisational governance for UKRI enables the leadership for this way of operating and bringing different parts of the system together. This is the challenge for STI organisations in leading transformative innovation approaches and require wider changes to UKRI's governance to ensure leadership can operate all of its internal levers to integrate research and innovation activity according to the needs of different types of Challenge. It also requires sufficient autonomy from BEIS, UKRI's parent department, to establish appropriate governance of Challenges across the wider innovation policy space (Boon and Edler, 2018; Kuhlmann and Rip, 2018). This includes empowerment to work in more experimental ways needed to enable transformative policies. Otherwise, the danger is these approaches are all rhetoric and a repackaging of traditional policies as "old wine [in] new bottles" (Wittmann et al., 2020 quoting Daimer et al., 2012 p.223).

When introduced in a particular context, missions don't operate in a vacuum but rather sit alongside other innovation 'frames', and as such can lead to contestation between different actors that can impact on the effectiveness of these programmes. The tensions between Challenge Directors hired from industry and senior public administrators in the policy establishment are an indication of this. At an operational level, a dominant civil service culture may not be helping UKRI's 'centre' get established in an effective way: "UKRI is a very large organization with very old-fashioned Public Sector management…. it's as if every transformation in industrial management that's happened since 1980 just hadn't occurred" (Interviewee 10). There may be value instead in introducing multi-actor teams to lead Challenges, including membership from both inside and outside of the organisation, who operate as advocacy coalitions for dealing with change and the conflicts that inevitably arise in the process (Sabatier and Weible, 2007).

5.2. Implications for transformative innovation policies

There are three sets of wider implications from our case study. First, as predicted by Wittmann et al. (2021), transformative innovation policies of the accelerator type might seem like a relatively low hanging fruit as policy makers can retool existing organisations and measures. As we can see in the case UKRI, such an approach brings with it potential pitfalls of incrementalism and 'mission washing': putting a new label on existing policies. Accordingly, policy actors should focus from the outset on developing new capacities for mission-oriented innovation policies.

Second, as predicted by Borrás, public leadership support, both at the level of ministries and agencies, seems key to ensure both wider buy-in into missions across the actors and better coordination along the way (Borrás, 2021). It does not seem to be enough to dedicate one agency to missions and let other sister agencies and parent ministries go on with business as usual. Transformative innovation policies including missions requires a division of labour amongst public actors and focusing on capacity development across those actors.

Third, managerial autonomy seems to provide key space for organisational learning and experimentation to take place. As shown by Breznitz and others (Breznitz and Ornston, 2013; Breznitz et al., 2018), somewhat peripheral public agencies have been able to build up more autonomy which has enabled them to implement transformational policies. Challenge Directors seem to play this role in UKRI, if to a more limited extent as they are challenged by incumbent practice and micromanaging. However, in Challenge Directors we can see emerging a kind of mission mystique that could help build a different kind of organisational identity for stronger capacities for transformational mission innovation policies.

Overall, the case study illustrates the gap between the promise of mission approaches and conceptual work on key characteristics and capacities needed, and then the reality of putting these frameworks into practice. This highlights a need for more in depth academic and practical understanding of policy capacities and capabilities. UKRI as a case study shows how capacity can serve also as useful lens in understanding policy effectiveness. This case study looked at a short period of implementation, a deeper study of all key public actors involved in mission-oriented innovation policies would yield wider range of insights. Also, as many countries are experiencing such policies, a larger comparative study of policy capacities and capabilities for transformative change could prove highly useful and relevant in policy practice.

CRediT authorship contribution statement

Julie McLaren: Conceptualization, Writing – original draft, Writing – review & editing, Methodology, Investigation, Formal analysis. **Rainer Kattel:** Conceptualization, Writing – review & editing, Writing – original draft, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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